

Effective Carbon Price

CES-Based Decomposition into Intensive and Extensive Margins

Draft Working Note

Notation and Units

Symbol	Definition	Units
f	Fuel index	—
s	Sector index	—
q_{fs}	Baseline energy consumption of fuel f in sector s	GJ
p_{fs}	Pre-tax price of fuel f in sector s	\$/GJ
e_f	Emission factor of fuel f	tCO ₂ /GJ
$E_{fs} = q_{fs} e_f$	Baseline emissions from fuel f in sector s	tCO ₂
$E = \sum_s \sum_f E_{fs}$	Total baseline emissions	tCO ₂
σ_s	CES substitution elasticity in sector s	—
η_s	Service demand elasticity in sector s	—
C_s	CES unit cost index in sector s	\$/GJ
$\bar{e}_s = \sum_f w_{fs} e_f$	Energy-share-weighted mean emission factor in sector s	tCO ₂ /GJ
$w_{fs} = q_{fs}/Q_s$	Energy share of fuel f in sector s	—
$Q_s = \sum_f q_{fs}$	Total energy in sector s	GJ
τ_{fs}^{int}	Intensive carbon price on fuel f in sector s	\$/tCO ₂
τ_s^{ext}	Extensive carbon price in sector s	\$/tCO ₂
τ^*	Total effective carbon price	\$/tCO ₂

CES-Derived Elasticities

From the CES with substitution elasticity σ_s and demand elasticity η_s :

Intensive margin elasticity (fuel switching, fuel f in sector s):

$$\varepsilon_{fs}^{\text{int}} = -\sigma_s (1 - w_{fs}) \quad (1)$$

This is the own-price elasticity of fuel f holding total sectoral energy constant: a 1% rise in the price of fuel f reduces its consumption by $\sigma_s(1 - w_{fs})\%$, redistributing energy to other fuels.

Extensive margin elasticity (demand reduction, sector s):

$$\varepsilon_s^{\text{ext}} = \eta_s \quad (2)$$

This is the elasticity of total sectoral energy demand to the CES cost index C_s .

Policy Price Decomposition

Any policy that changes fuel prices by Δp_{fs} can be decomposed.

Extensive component (average cost increase in sector s):

$$\tau_s^{\text{ext}} = \frac{\sum_f w_{fs} \Delta p_{fs}}{\bar{e}_s} \quad (3)$$

This converts the energy-share-weighted average price increase into carbon-price-equivalent units by dividing by the mean emission factor. For a pure carbon tax at rate τ , this gives $\tau_s^{\text{ext}} = \tau$.

Intensive component (relative price signal for fuel f):

$$\tau_{fs}^{\text{int}} = \frac{\Delta p_{fs} - \sum_{f'} w_{f's} \Delta p_{f's}}{e_f - \bar{e}_s} \quad (4)$$

This is the deviation of fuel f 's price change from the sectoral average, normalised by the deviation of its emission factor from the mean. For a pure carbon tax at rate τ , this also gives $\tau_{fs}^{\text{int}} = \tau$ for all fuels. For a revenue-neutral feebate, $\tau_s^{\text{ext}} \approx 0$ while τ_{fs}^{int} is the feebate's implicit carbon price.

Emissions Reduction by Fuel and Sector

The linearised emissions reduction from fuel f in sector s :

$$\Delta E_{fs} = E_{fs} \left[\underbrace{\sigma_s (1 - w_{fs}) \frac{\tau_{fs}^{\text{int}} (e_f - \bar{e}_s)}{p_{fs}}}_{\text{intensive margin}} + \underbrace{|\eta_s| \frac{\tau_s^{\text{ext}} \bar{e}_s}{C_s}}_{\text{extensive margin}} \right] \quad (5)$$

The first term captures fuel switching: fuels with above-average emission factors ($e_f > \bar{e}_s$) see consumption fall; fuels with below-average emission factors see consumption rise (through the cross-price elasticity from the CES). The second term captures demand reduction: total energy demand in the sector falls in proportion to the cost increase.

Total emissions reduction:

$$\Delta E = \sum_s \sum_f \Delta E_{fs} \quad (6)$$

Benchmark: \$1 Carbon Tax

Under a uniform carbon tax of \$1/tCO₂, every fuel's price rises by $\Delta p_{fs} = e_f$, giving $\tau_{fs}^{\text{int}} = 1$ and $\tau_s^{\text{ext}} = 1$ for all f, s . The benchmark emissions reduction is:

$$\Delta E^{\text{bench}} = \sum_s \sum_f E_{fs} \left[\sigma_s (1 - w_{fs}) \frac{e_f - \bar{e}_s}{p_{fs}} + |\eta_s| \frac{\bar{e}_s}{C_s} \right] \quad (7)$$

Effective Carbon Price

The effective carbon price τ^* is the uniform carbon tax that produces the same total emissions reduction as the actual policy mix:

$$\tau^* = \frac{\Delta E}{\Delta E^{\text{bench}}} = \frac{\sum_s \sum_f E_{fs} \left[\sigma_s (1 - w_{fs}) \frac{\tau_{fs}^{\text{int}} (e_f - \bar{e}_s)}{p_{fs}} + |\eta_s| \frac{\tau_s^{\text{ext}} \bar{e}_s}{C_s} \right]}{\sum_s \sum_f E_{fs} \left[\sigma_s (1 - w_{fs}) \frac{e_f - \bar{e}_s}{p_{fs}} + |\eta_s| \frac{\bar{e}_s}{C_s} \right]} \quad (8)$$

Or equivalently, defining weights:

$$\tau^* = \frac{\sum_s \left[\tau_s^{\text{ext}} \cdot \Omega_s^{\text{ext}} + \sum_f \tau_{fs}^{\text{int}} \cdot \Omega_{fs}^{\text{int}} \right]}{\sum_s \left[\Omega_s^{\text{ext}} + \sum_f \Omega_{fs}^{\text{int}} \right]} \quad (9)$$

where:

$$\Omega_s^{\text{ext}} = |\eta_s| \frac{\bar{e}_s}{C_s} \sum_f E_{fs} = |\eta_s| \frac{\bar{e}_s E_s}{C_s} \quad (10)$$

$$\Omega_{fs}^{\text{int}} = \sigma_s (1 - w_{fs}) \frac{(e_f - \bar{e}_s)}{p_{fs}} E_{fs} \quad (11)$$

The Ω weights are *properties of the economy*, not of the policy. They are computed once from the energy balance, fuel prices, emission factors, and CES elasticities. The policy enters only through τ_{fs}^{int} and τ_s^{ext} .

Compact Form (Suppressing Indices)

Dropping fuel and sector subscripts and writing the sums implicitly:

$$\tau^* = \frac{\sum E \left[\sigma(1-w) \frac{\tau^{\text{int}}(e - \bar{e})}{p} + |\eta| \frac{\tau^{\text{ext}} \bar{e}}{C} \right]}{\sum E \left[\sigma(1-w) \frac{e - \bar{e}}{p} + |\eta| \frac{\bar{e}}{C} \right]} \quad (12)$$

where $\sum$ runs over all fuel–sector pairs, $\bar{e}$ and C are sector-level, and all other quantities are fuel–sector-level.

Inputs and Computation

Given (from energy balance and price data):

- q_{fs} — energy consumption (GJ) for each fuel f in sector s
- p_{fs} — pre-tax fuel price (\$/GJ)
- e_f — emission factor (tCO₂/GJ)

Given (from CES calibration):

- σ_s — substitution elasticity per sector
- η_s — demand elasticity per sector

Given (from policy description):

- Δp_{fs} — price change on each fuel from the policy (\$/GJ)

Compute:

1. $w_{fs} = q_{fs} / \sum_{f'} q_{f's}$
2. $\bar{e}_s = \sum_f w_{fs} e_f$
3. $E_{fs} = q_{fs} e_f$
4. $C_s = \left(\sum_f \alpha_{fs} p_{fs}^{1-\sigma_s} \right)^{1/(1-\sigma_s)}$ from CES calibration
5. $\tau_s^{\text{ext}} = \sum_f w_{fs} \Delta p_{fs} / \bar{e}_s$

6. $\tau_{fs}^{\text{int}} = \left(\Delta p_{fs} - \sum_{f'} w_{f's} \Delta p_{f's} \right) / (e_f - \bar{e}_s)$
7. Ω_s^{ext} and Ω_{fs}^{int} from the formulae above
8. $\tau^* = \frac{\sum_s \left[\tau_s^{\text{ext}} \cdot \Omega_s^{\text{ext}} + \sum_f \tau_{fs}^{\text{int}} \cdot \Omega_{fs}^{\text{int}} \right]}{\sum_s \left[\Omega_s^{\text{ext}} + \sum_f \Omega_{fs}^{\text{int}} \right]}$

Output: τ^* (\$/tCO₂) — the effective carbon price.

Special Cases

Pure carbon tax at rate τ : $\tau_{fs}^{\text{int}} = \tau_s^{\text{ext}} = \tau$ for all f, s , so $\tau^* = \tau$. The formula is self-consistent.

Revenue-neutral feebate: $\tau_s^{\text{ext}} \approx 0$, so only the intensive terms survive. The effective carbon price is the feebate rate scaled by the ratio of intensive-margin effectiveness to total carbon-tax effectiveness.

Uniform energy tax (same \$/GJ on all fuels): $\tau_{fs}^{\text{int}} = 0$ for all f , so only the extensive terms survive. The effective carbon price reflects only the demand reduction channel.

Single sector, single fuel: σ is irrelevant (no switching possible), τ^* reduces to the extensive component only, and equals the fuel price increase divided by the emission factor.